Lesson notes · for you

If this, then that

Programming · Level 1 · Lesson 3

Time	Materials
20 min	Paper and coloured pencils; A dice (optional); A phone (optional)

Learning goals

- Say what a decision is.
- Use AND and OR in simple rules.
- Test a rule in every case.

Activity: Simon says IF

1. Warm-up (3 min): Listen to the rule: “IF you have a brother, THEN stand up, ELSE touch your nose.” Do what it says.
2. Play (7 min): Now rules have AND or OR: “IF you like football AND you have a sister, THEN clap.” Who should clap?
3. Make (7 min): Write your own rule card with IF, THEN, ELSE and one AND or OR.
4. Test (3 min): Swap cards with a partner. Check the rule in all four cases: does it always say what to do?

Keep the rules kind: no rule should single out one person.

Discussion

- What is the difference between AND and OR?
- Why test every case?
- Which rules at home or at school use AND or OR?

Check yourself

- Write one rule with AND and one with OR.

Try it this week
Play “Simon says IF” with your family: “IF you are wearing something red, THEN stand up, ELSE sit down.” Then try a rule with AND.

Worksheet

If this, then that

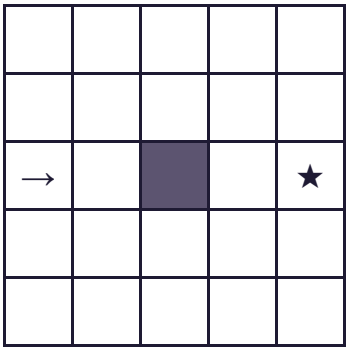
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Name _____ Class _____ Date _____

1 **The alarm rings IF the door OR the window opens. Only the window opens. Does it ring?**

◯ Yes ◯ No

2 **A box is in the robot's way. You can go around it at the top or at the bottom: you decide! Use 7 cards at most.**



Cut out cards (↑ forward, ↶ turn left, ↷ turn right) and lay out a program that takes the robot to the ★. Write it here:

3 **With AND, the lamp goes on when only one switch is on.**

◯ True ◯ False

4 **Match each word to what it means in a rule.**

IF	• • • •	What to do when it is not true
ELSE	• • • •	One is enough
AND	• • • •	Checks if something is true
OR	• • • •	Both must be true

5 **The door alarm rings IF the door opens OR the window opens. When does it ring?**

☐ Door shut, window open ☐ Door open, window shut ☐ Both open ☐ Both shut

✎ **Write one IF … ELSE rule from your home or school. Then write a version with AND or OR.**

Check your answers

If this, then that

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1 **Yes**

With OR, one yes is enough. The window is open, so the alarm rings.

2 **Any program that takes the robot to the ★ with 7 cards or fewer.**

Turn left, forward, turn right, forward 4 times, turn right, forward: six cards over the top.

3 **False**

AND needs both switches on. With only one, the lamp stays off.

4 **IF — Checks if something is true**
ELSE — What to do when it is not true
AND — Both must be true
OR — One is enough

IF checks, ELSE says what to do otherwise, AND needs both, OR needs only one.

5 **Door open, window shut; Door shut, window open; Both open**

With OR, it rings when at least one is open.